

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details: Batch Number 13

Start date: 03/02/2026

Name: Soham Badgujar

Internship Duration: 8 Weeks

1. Project Title

Global Income Intelligence Platform: Interactive Data Analytics and Machine Learning Dashboard.

2. Project Objective

The objective of this project is to develop an interactive data analytics platform that analyzes global income distribution and inequality trends. The system aims to provide meaningful insights using visualization techniques and predictive models.

The project focuses on:

- understanding global income inequality
- performing exploratory data analysis
- building interactive dashboards
- integrating machine learning for prediction
- enabling real-time insights generation

3. Project description in detail

The Global Income Intelligence Platform is a web-based analytical system developed to study income distribution patterns across countries.

The project was implemented using:

- Python for programming
- Pandas for data processing
- Streamlit for web interface
- Plotly and Matplotlib for visualization
- Scikit-learn for machine learning
- Microsoft Power BI for dashboard integration

Approach

1. Data Collection and Cleaning
2. Exploratory Data Analysis (EDA)
3. Visualization Development
4. Machine Learning Model Implementation
5. Dashboard Integration

Real-World Impact

- helps understand economic inequality trends
- supports policy-level insights
- enables data-driven decision making
- provides predictive analytics for future trends

From a technological perspective, the project integrates data analytics tools with modern web development frameworks, creating a complete end-to-end solution. It demonstrates how raw data can be transformed into meaningful insights and delivered through an interactive digital platform.

In real-world scenarios, this project can be useful for researchers, students, and policymakers who want to study economic inequality and make data-driven decisions. It simplifies complex datasets into intuitive visuals, making analysis more accessible and impactful.

4. Timeline Overview

Week Activities	Planned	Activities Completed
Week 1	Basics of Python & Data Analysis	Learned Python, Pandas
Week 2	Data Cleaning	Handled missing values, preprocessing
Week 3	EDA	Generated insights & visualizations
Week 4	Power BI Dashboard	Created interactive BI dashboard
Week 5	Streamlit App Development	Built UI and integrated modules
Week 6	Advanced Visualization	Added charts & maps
Week 7	Machine Learning	Implemented regression models
Week 8	Final Integration & Deployment	Completed project & testing

5a. Key Milestones

Milestone	Description	Date Achieved
Project Kickoff	Data collection started	03-02-2026
Prototype	Initial dashboard created	20-02-2026
Mid-Term Review	Progress evaluation	27-02-2026
Final Submission	Web portal completed	20-03-2026
Presentation	Final demo	26-03-2026

5b. Project execution details

This project was developed in multiple structured phases:

Phase 1 – Data Processing

What i actually did:

In this phase, I cleaned and structured the dataset to ensure accuracy. Handling missing values and duplicates was important to avoid incorrect analysis. The raw dataset was first loaded and processed using Pandas.

Steps involved:

- Data Loading Imported dataset using `pd.read_csv()`
- Handling Missing Values Identified null values using `.isnull()`
- Filled or removed missing data using:
 - `fillna()`
 - `dropna()`
- Duplicate Removal Checked duplicates using `.duplicated()`
- Removed using `.drop_duplicates()`
- Data Type Correction Converted columns into proper formats (int, float, category)
- Feature Selection Selected relevant columns for analysis

Phase 2 – Visualization

What i did:

Visualization helps transform raw data into insights. I used interactive charts so users can explore data dynamically. Created multiple visual representations using:

Plotly → interactive charts

Matplotlib → basic/static plots

Types of charts implemented:

Histogram → data distribution

Scatter Plot → relationship between variables

Line Chart → trend over time

Box Plot → outliers

Heatmap → correlation

Phase 3 – Dashboard Development

What i built:

I developed a modular dashboard where each section performs a specific analytical task, making the system user-friendly and scalable. A complete web interface using Streamlit.

Modules Created:

1. Executive Dashboard

dataset overview

KPI metrics

summary charts

2. Dataset Explorer

- missing values
- duplicates
- statistics

Purpose:

“Performs exploratory data analysis (EDA)”.

3. Visualization Panel

- user selects variables
- generates charts dynamically

Purpose:

“Allows interactive data exploration”.

4. Country Analysis

- country-specific insights
- trend analysis

Purpose:

“Deep analysis for individual country”.

UI Features:

- sidebar navigation
- dropdown filters
- dynamic charts
- responsive layout

Phase 4 – Machine Learning

What i implemented:

I implemented multiple models and compared their performance. Random Forest provided better accuracy due to its ensemble nature. Used Scikit-learn to build predictive models.

Models Used:

1. Linear Regression

simple prediction model

assumes linear relationship

Use: baseline model

2. Decision Tree

rule-based model

splits data into branches

Use: handles non-linear patterns

3. Random Forest

multiple decision trees

higher accuracy

Use: best performing model

Steps involved:

- feature selection
- train-test split
- model training
- prediction
- evaluation using R^2 score

Outputs shown:

- actual vs predicted graph
- feature importance
- model comparison

Phase 5 – Integration

What i did:

In the final phase, I integrated all modules into a single platform, combining data analysis, visualization, and machine learning into one system. Combined all components into one unified platform.

Integration Includes:

- Power BI Integration
- Using Microsoft Power BI
- embedded dashboard using iframe
- interactive filters

Purpose:

“Adds enterprise-level visualization”

Module Integration

- EDA + Visualization + ML combined
- smooth navigation between sections

Final Output

- single web application
- fully interactive
- real-time analysis

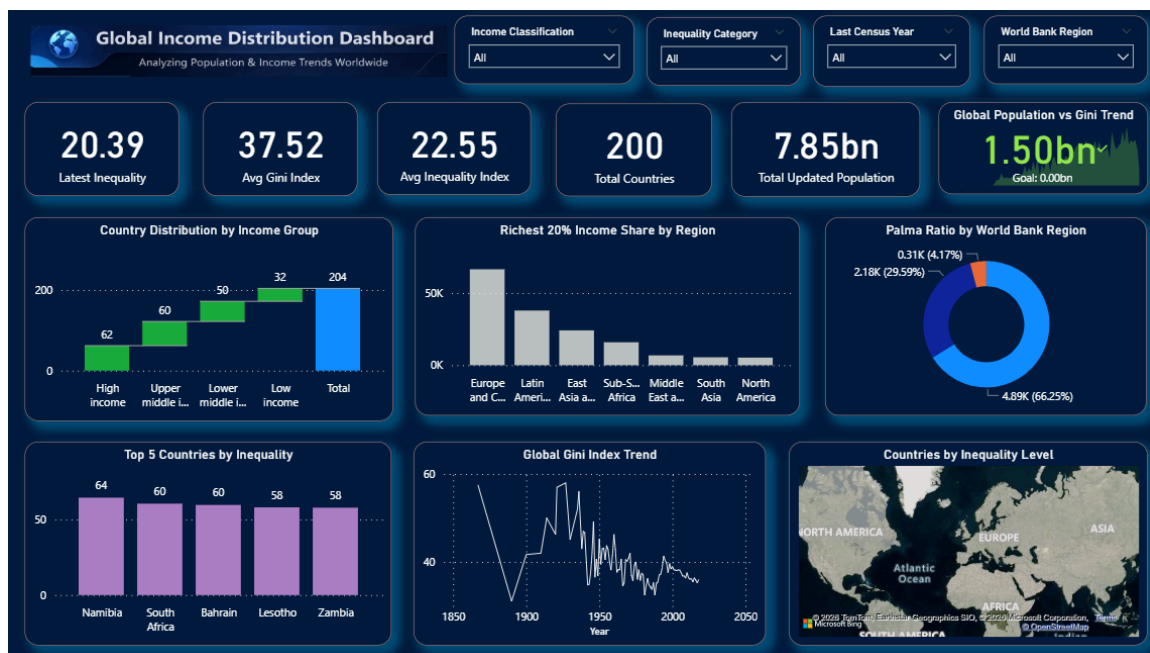
The project was executed in five phases. First, data processing was done using Pandas to clean and structure the dataset. Next, visualization was performed using Plotly and Matplotlib to identify patterns. Then, a modular dashboard was developed using Streamlit with multiple analytical sections. After that, machine learning models like Linear Regression, Decision Tree, and Random Forest were implemented for prediction. Finally, all components were integrated, including embedding a Power BI dashboard, resulting in a complete interactive analytics platform

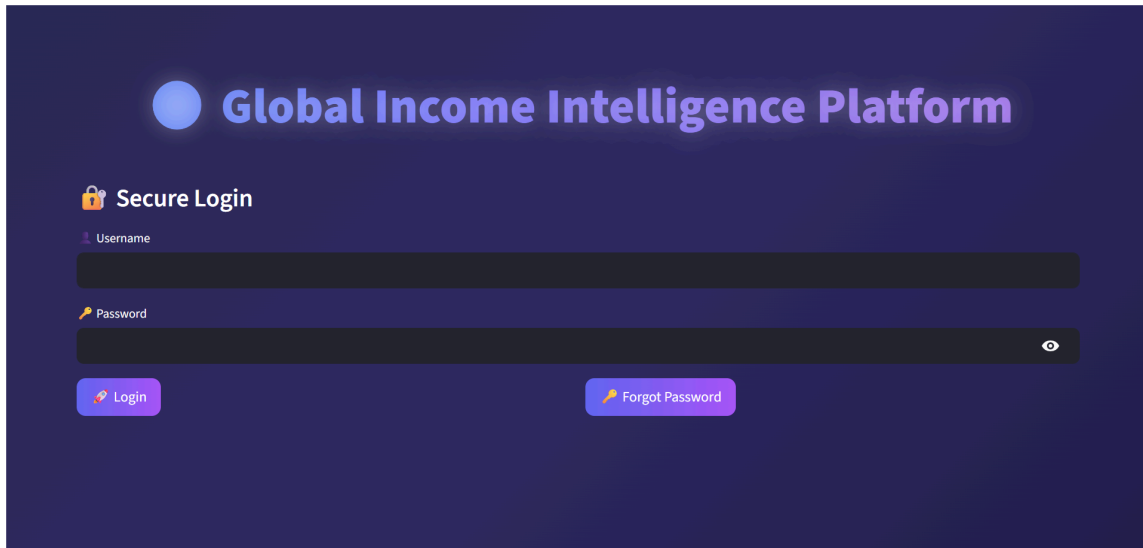
6. Snapshots / Screenshots

Dataset After Cleaning and Merging

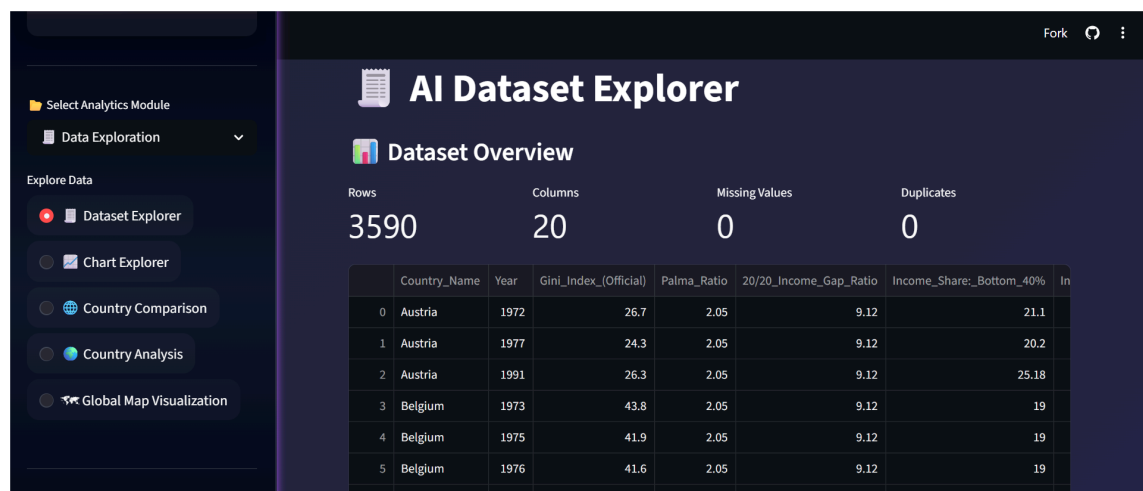
	Gini Index (Official)	Palma Ratio	20/20 Income Gap Ratio	Income Share: Bottom 40%	Income Share: Poorest 20%	Income Share: Second Quintile	Income Share: Middle Quintile	Income Share: Fourth Quintile
2008	29	2.05	4.33	22	9	13	17	
2012	33	2.05	4.11	23	9	14	17	
2017	31	2.05	4.44	22	9	13	16	
1996	27.01	0.93	3.95	22.85	9.15	13.7	17.74	
2002	31.74	1.22	4.83	20.93	8.35	12.58	16.49	
2005	30.6	1.14	4.66	21.3	8.4	12.9	17.02	
2008	29.98	1.11	4.4	21.94	8.87	13.07	16.84	
2012	28.96	1.04	4.27	22.02	8.85	13.17	17.34	
2012	27.62	0.99	3.98	23.02	9.36	13.66	17.47	
1988	40.19	1.9	7.21	17.28	6.54	10.74	14.93	
1995	35.33	1.45	6.11	18.52	6.97	11.55	16.24	
2001	26.46	2.05	9.12	22.02	6.53	11.16	15.48	
2002	26.76	2.05	9.12	22.02	6.53	11.16	15.48	
2003	27.21	0.99	3.78	23.7	9.9	13.8	17.1	
2016	27.96	1	3.91	23.1	9.7	13.4	16.8	
1995	45	2.05	9.12	22	6.53	11.16	15.48	
2001	51.96	3.65	17.65	11.02	3.18	7.84	12.71	
2009	42.72	2.15	8.94	15.05	5.43	9.62	14.46	
1976	35.2	2.05	9.12	17.3	7	11.16	15.48	
1977	40.3	2.05	9.12	17.3	6.4	11.16	15.48	
1978	43.4	2.05	9.12	17.3	5.6	11.16	15.48	
1979	41.3	2.05	9.12	17.3	5.9	11.16	15.48	
1982	40.2	2.05	9.12	14.6	6	11.16	15.48	
1983	40.4	2.05	9.12	14.6	5.5	11.16	15.48	
1985	39.8	2.05	9.12	14.6	6.3	11.16	15.48	
1993	40	2.05	6.77	18.1	7.4	10.7	13.6	
1999	44.7	2.61	8.06	16.2	6.8	9.4	12.3	
1961	41.9	2.25	7.41	17.3	7	10.3	13.2	
1963	37.3	1.71	6.72	17.8	6.8	11	15.2	
1965	35.3	1.49	6.67	18.1	6.4	11.7	16.3	
1969	37.3	1.69	7.24	17.3	6.2	11.1	15.7	
1970	36.1	1.53	6.89	18	6.3	11.7	16.2	
1972	35.3	1.51	6.83	18.3	6.3	12	16.6	
1974	34.5	1.38	5.81	19.1	7.2	11.9	16.5	
1975	36.1	1.54	7.27	17.3	5.9	11.4	16.5	
1981	42.7	2.29	9.56	14.6	5.2	9.4	14.3	
1988	45.5	2.51	11.5	13.5	4.4	9.1	14	
1989	47.6	2.83	12.83	12.7	4.1	8.6	13.3	
1990	50.1	2.18	13.1	12.4	4.2	8.2	13.6	

Power BI Dashboard



Authenticated Area

The login page features a dark blue background with a glowing blue circle icon. The title 'Global Income Intelligence Platform' is displayed in a light blue font. Below the title, there is a 'Secure Login' section with a lock icon. It includes two input fields: 'Username' and 'Password'. The 'Password' field has a toggle icon (an eye) to show or hide the password. At the bottom of the login section, there are two buttons: 'Login' and 'Forgot Password'.

Home Page**AI Dataset Explorer**

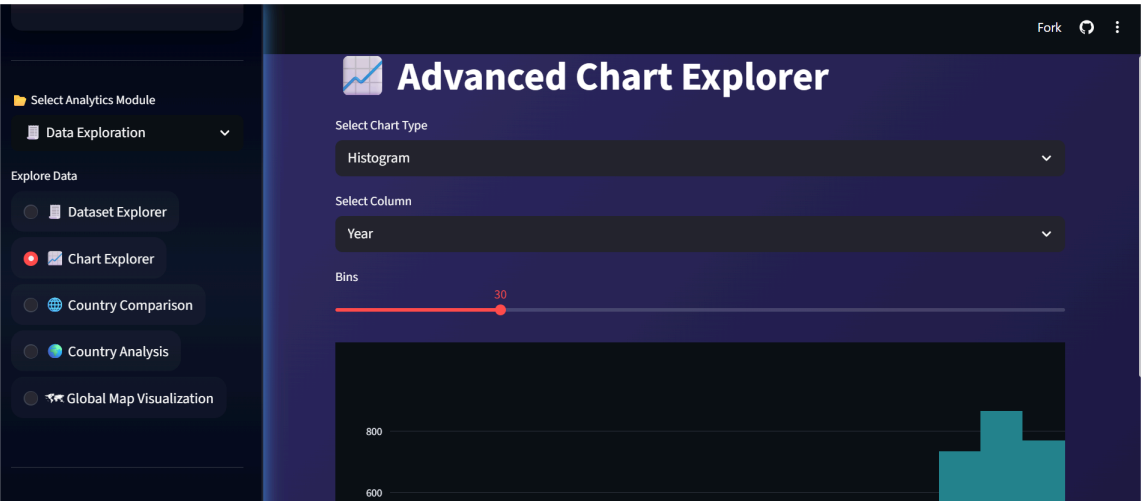
The AI Dataset Explorer interface shows a sidebar with 'Select Analytics Module' (Data Exploration) and 'Explore Data' (Dataset Explorer, Chart Explorer, Country Comparison, Country Analysis, Global Map Visualization). The main content area is titled 'AI Dataset Explorer' and 'Dataset Overview'. It displays summary statistics: Rows (3590), Columns (20), Missing Values (0), and Duplicates (0). Below this is a table of data.

	Country_Name	Year	Gini_Index_(Official)	Palma_Ratio	20/20_Income_Gap_Ratio	Income_Share:_Bottom_40%	In
0	Austria	1972	26.7	2.05	9.12	21.1	
1	Austria	1977	24.3	2.05	9.12	20.2	
2	Austria	1991	26.3	2.05	9.12	25.18	
3	Belgium	1973	43.8	2.05	9.12	19	
4	Belgium	1975	41.9	2.05	9.12	19	
5	Belgium	1976	41.6	2.05	9.12	19	
6	Belgium	1977	40.6	2.05	9.12	19	

Power BI Integrated Dashboard



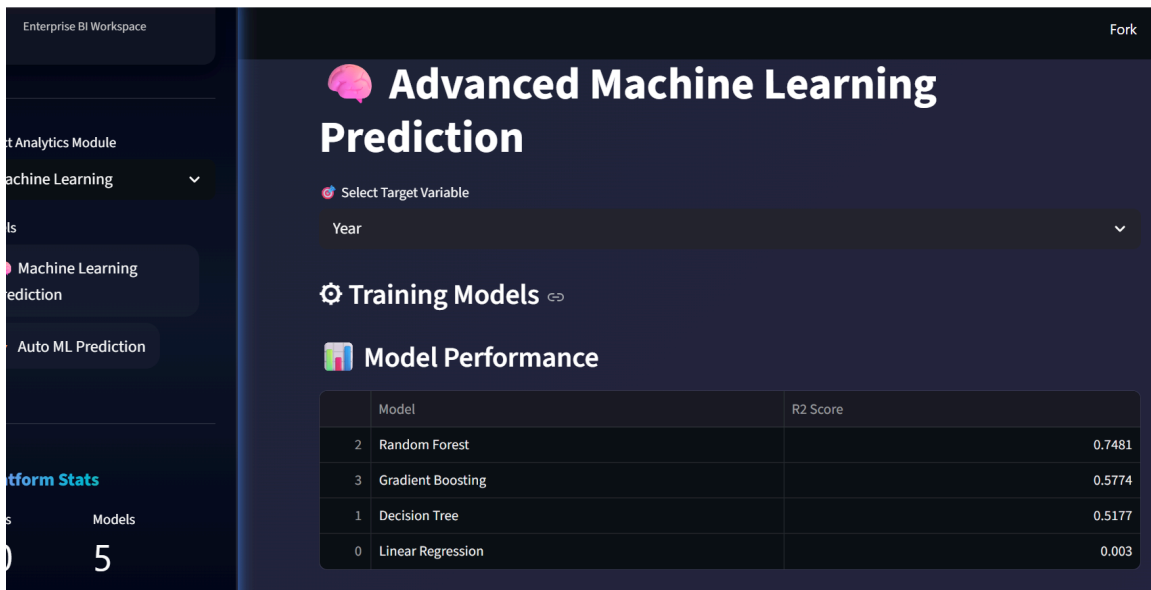
Advanced Chart Explorer



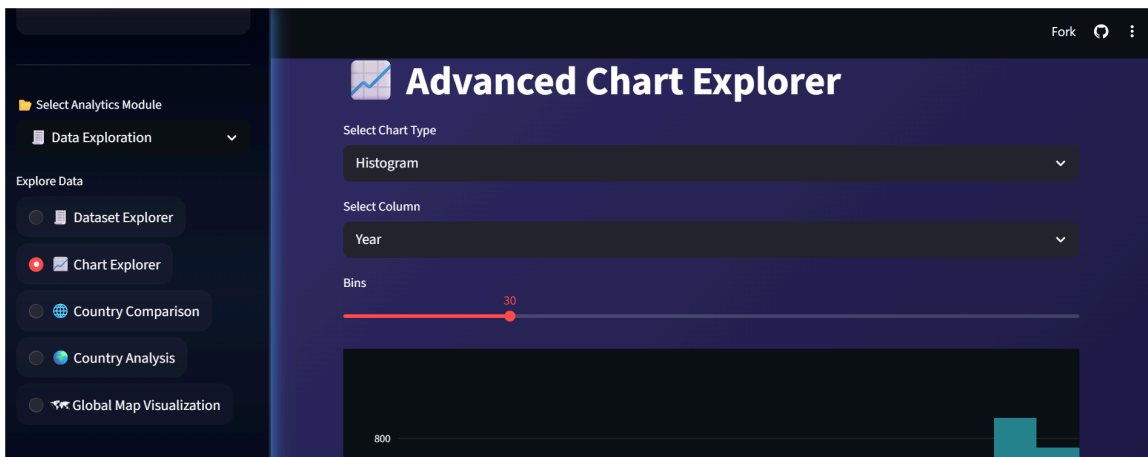
Country Comparison Dashboard



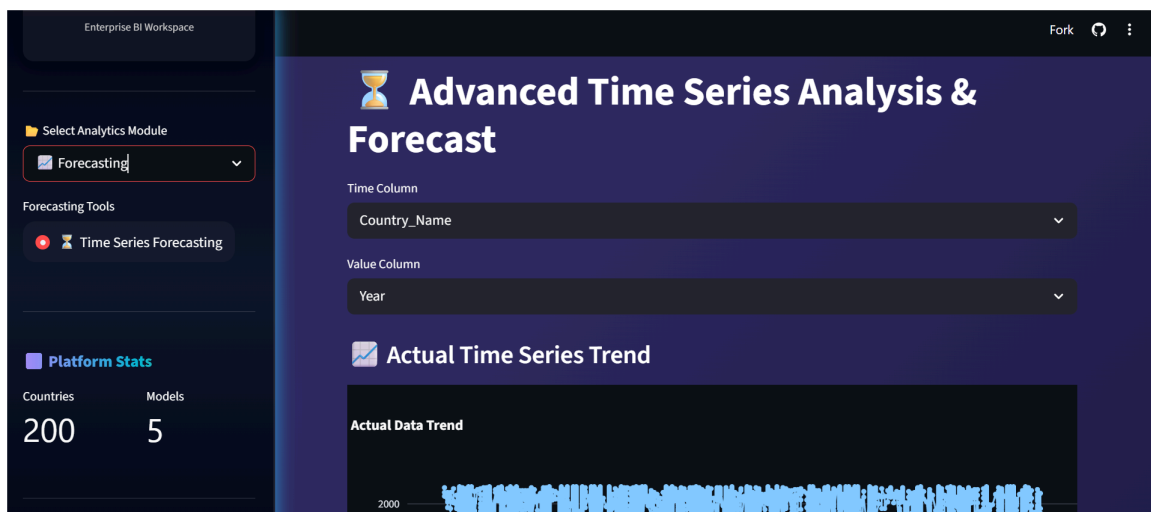
Advanced Machine Learning Prediction



Advanced Chart Explorer



Advanced Time Series Analysis & Forecast



7. Challenges Faced

1. Data Quality Issues

Missing values and inconsistent data

Solved using preprocessing techniques

2. Streamlit UI Design

Difficulty in creating professional UI

Solved using custom CSS and layout

3. Machine Learning Accuracy

Low model performance initially

Improved using feature selection and tuning

4. Deployment Issues

Configuring deployment platform

Solved by correct setup and commands

8. Learnings & Skills Acquired

Technical Skills

- Data Analysis using Pandas
- Data Visualization
- Machine Learning basics
- Dashboard Development
- Web App Deployment

Tools Learned

- Python
- Streamlit
- Power BI
- Scikit-learn

Soft Skills

- Problem solving
- Analytical thinking
- Debugging
- Project management

9. Testimonials from team

This internship provided valuable hands-on experience in real-world data analytics. The project helped in understanding how data can be transformed into meaningful insights using visualization and machine learning.

10. Conclusion

The internship successfully enhanced my understanding of data analytics, visualization, and machine learning.

The project demonstrated how analytical tools can be used to study global income inequality and generate predictive insights.

It also strengthened my practical skills and prepared me for real-world data science roles.

11. Acknowledgements

I would like to thank Infosys Springboard for providing this internship opportunity.

I also express gratitude to my mentors and team members for their continuous support and guidance throughout the project.